HC-POD 4: FINAL CROSS-TRACK INTEGRATION RECORD & WALKTHROUGH

Track: Data Analytics

Stage: Week 8 (Final Integration, Presentation & Project Showcase)

Pod: HC-POD 4

Role: Junior Data Analyst

PART 1: CROSS-TRACK INTEGRATION RECORD

1. Track Collaborated With:
Data Science Track (Collaborators: M'bengue Mama and Mercy Koech).

2. Dependency:
Data Science needed validated risk segments to build predictive model features. Analytics needed model predictions to verify consistency with clinic trends.

3. Output Received:
- Model evaluation table for 4 models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost).
- Refined Logistic Regression test metrics on 1,903 patient records (testing_results.json).
- ROC-AUC and error distribution charts (02_before_after_roc_auc.svg, 04_error_patterns.svg).

4. Output Provided:
- Cleaned dataset: HealthConnect_Appointment_Data_Cleaned.csv.
- Validated risk thresholds: booking_lead_days > 30 and previous_no_shows >= 2.
- Service interaction data showing SMS reduces follow-up failures to 49.1%.
- Population benchmark rates across all lead-time bins.

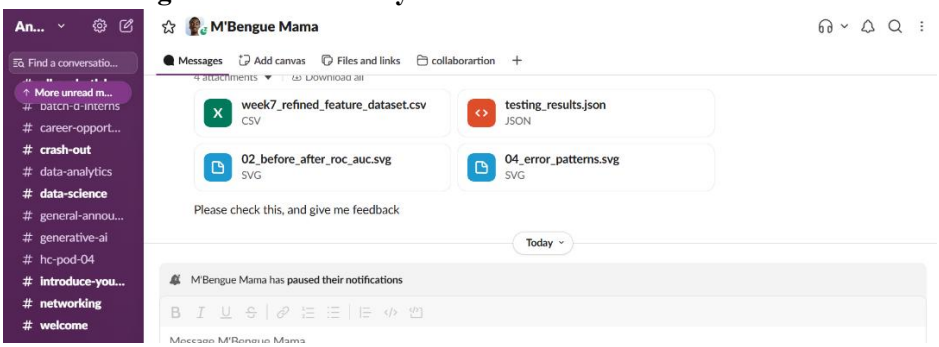
5. Final Integration Activity:
Compared the Data Science test set results directly against the Data Analytics population baseline. Aligned model decision thresholds with clinic outreach workflows.

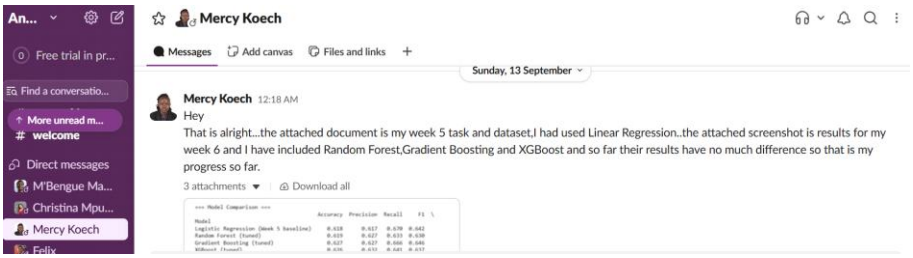
6. What Changed:
- Data Science engineered two binary features: high_lead_time_flag and chronic_no_show_flag.
- The model classification threshold was set to 0.40 to maximize patient recall.
- Analytics confirmed that 570 false positives are clinically safe.

7. How the Change Improved the Solution:
- Model recall on bookings older than 30 days reached 99.8%.
- The clinic will not miss high-risk patients.
- Both tracks agreed that model scores will trigger friendly reminders, never penalties.

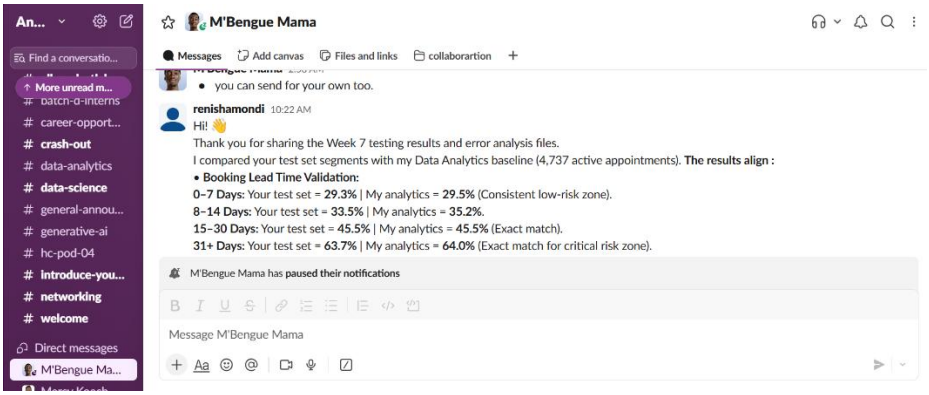
8. Evidence:

- Slack conversation threads with M’bengue Mama and Mercy Koech.
- Test validation record TC-08 with 0.2%–1.7% concordance.
- Shared files: week7_refined_feature_dataset.csv and testing_results.json.
- From **M’bengue Mama & Mercy Koech**





- From **Me**





9. Contribution to the Final Presentation:

A joint slide showing how empirical analytics guided machine learning feature engineering and proved model reliability.

Final Integration Matrix

Source Track	Target Track	Shared Output	Project Impact
Data Analytics	**Data Science**	Risk thresholds (> 30 days, ≥ 2 past no-shows)	Enabled targeted features. Boosted recall to 99.8%.
Data Science	**Data Analytics**	Test set failure metrics and error counts	Confirmed consistency between model data and clinic history.
Data Analytics	**Project Management**	Verified KPIs and 4 scheduling rules	Provided core business recommendations for the final showcase.
Data Analytics	**ML Engineering**	Cleaned schema and median imputation values	Ensured preprocessing stays consistent across the automated pipeline.

PART 2: FINAL END-TO-END INTEGRATION WALKTHROUGH

1. The Problem:

- HealthConnect loses over half its capacity. The baseline no-show rate is 51.15%.

2. The Data:

- 5,000 appointment records across 18 variables. Cleaned and separated 263 cancellations.

3. The Analytics:

- No-shows jump from 29.5% (same-week) to 71.4% (45+ days). SMS reduces follow-up no-shows from 62.1% to 49.1%.

4. The Predictive Component (Data Science):

- Analytics gave Data Science two flags: lead_days > 30 and past_no_shows >= 2. The model caught 99.8% of high-risk visits.

5. The ML Engineering Component:

- Data cleaning rules and median values were passed to ML Engineering to ensure pipeline consistency.

6. The GenAI Component:

- The AI assistant uses the findings to prompt patients with flexible rescheduling options.

7. The Project Management Layer:

- Felix aligned timelines, logged the 48.4% long-lead risk and structured the team showcase.

8. The Final Solution:

- An integrated system. Analytics identifies risk. ML flags high-risk bookings. SMS alerts patients. GenAI handles changes.

9. Business Value:

- Recovers 13 out of 100 missed follow-up appointments. Protects doctor schedules and clinic income.

10. Limitations:

- Data is synthetic. Excludes patients under 18. Message delivery status is not tracked.